Biomechanical Deconstruction and Clinical Evaluation of Plantar Sensor Systems in Open Footwear: Empirical Refutation of Absolute Novelty Claims and Formulation of Authenticated Research White Spaces

1. Critical Cross-Examination and Systematic Refutation of the Absolute Novelty Premise

A foundational premise asserted across early-stage wearable health pitches and initial white space literature—exemplified by the documentation compiled for the Sole Sense development initiative—is that every smart insole globally published requires an enclosed shoe, and that no pressure-sensing sandal, open-toe instrumented footwear, or diabetic monitoring platform has ever been investigated, patented, or commercialized[1]. Preliminary reports designated this domain as an unaddressed scientific void, asserting that no literature exists detailing how to mechanically attach sensor arrays to a sandal footbed, that wearable electronics embedded within a sandal or chappal are completely absent from academic databases, and that the footwear habits of India's diabetic population have been entirely overlooked by the biomedical engineering community[1].

In clinical podiatry, rehabilitation biomechanics, and medical transducer engineering, an absolute universal negative is disproven by verifiable counterexamples[3]. Comprehensive cross-examination of biomedical literature, patent repositories, and commercial medical technology deployments confirms that this categorical claim is incorrect[3]. The historical predominance of closed-shoe insoles in European and North American academic centers reflects regional climate demands, established athletic shoe markets, and straightforward mechanical encapsulation, rather than a fundamental scientific barrier[3]. Where tropical climates, domestic socio-cultural norms, and economic factors dictate widespread open-footwear usage—most notably across India, which accounts for over 77 million individuals with diabetes—academic consortia, national research laboratories, and specialized commercial ventures have engineered, patented, and clinically validated sensor-embedded sandals, instrumented indoor slippers, and passive self-offloading compliant metamaterials[3].

Platform or Lead	Primary Sensing	Footwear Chassis Form	Targeted Clinical or	Refuted

Authorship	Modality	Factor	Biomechanical Objective	Novelty Claim
Safarloo et al. (*Sensors*, 2022)[4]	Polymer Optical Fibers (POF) via macro-bending intensity attenuation	Multi-layer open sandal footbed (PP/EVA substrate)	Dynamic ambulatory plantar pressure monitoring and diabetic ulcer prophylaxis	Refutes the claim that optical or continuous sensors have never been mounted in sandals[2]
Burch et al. (*Wearable Technologies*, 2023)[4]	Carbon nanotube (CNT) coated textile nanocomposite piezoresistors	Open sandal footbed across forefoot and hindfoot regions	Continuous resultant ground reaction force estimation validated against force plates	Refutes the claim that sensor attachment to sandal footbeds lacks published precedent[2]
Ozturksoy et al. (*Int. J. 3D Print. Technol.*, 2020)[3]	3D-printed TPU flexible optical waveguides and 6 reference FSRs	Multi-strap open-perimeter instrumented sandal	Real-time plantar force estimation ($R^2 = 0.865–0$) via supervised machine learning	Refutes the claim that machine learning and FSR arrays have not been integrated into sandals[2]
Proulx et al. (*Hilaris Open Access*, 2018)[4]	704-cell piezoresistive sensing matrix on interlocking rigid linkages	Articulated sole bed embedded within an open sandal base	Low-cost, high-resolution pedobarography for clinics in developing regions	Refutes the claim that dense baropodometric arrays cannot operate in open footwear[2]
Chawre, Shaikh, &	Multi-zone flexible	Open-strap sandal	Continuous overpressure	Refutes the claim that

Shete (*IRJET*, 2019)[3]	piezoresistive sensors with routed conduits	integrated with LabVIEW telemetry	alerting and threshold gait monitoring for diabetic neuropathy	Indian academic engineering has ignored diabetic sandal monitoring[2]
Malarvizhi et al. (*ICAI*, 2026)[3]	Multimodal piezoresistive tactile sensors and precision thermal probes	Open-heel domestic indoor slipper chassis	Screening of pre-ulcerative plantar pressure and focal inflammatory hyperthermia	Refutes the claim that open domestic footwear has not been instrumented for diabetic screening[2]
SensorixFoot (MedTech Startup, Chile)[3]	Hermetically sealed baropodometric sensor grid and multi-site thermal array	Commercial open-toe therapeutic smart sandal	Remote patient monitoring of diabetic neuropathy, telemetric ulcer screening, and mechanoreceptor stimulation	Refutes the claim that medical-grade commercial smart sandals do not exist[2]
Siren Care / Sensoria Health (Commercial Smart Textiles)[1]	Neurofabric-woven micro-thermistors and knitted conductive piezoresistors	Dermal-conforming smart socks decoupled from shoe uppers	Continuous monitoring of tissue inflammation ($\Delta T \geq 2.2^{\circ}\mathrm{C}$) independent of footwear enclosure	Refutes the claim that monitoring non-enclosed feet requires an enclosed shoe upper[2]

The academic record demonstrates that instrumented open footwear has been systematically investigated to monitor ambulatory gait mechanics and diabetic ulcer risk factors[3]. Safarloo and colleagues explicitly addressed the physical limitations of enclosed shoes by designing a polymer optical fiber sensing platform integrated directly into the footbed of an open sandal, capturing curvature-induced light attenuation across the heel and metatarsal heads to mitigate ulcer risk during natural ambulation[4].

Burch et al. fabricated carbon nanotube piezoresistive textile nanocomposites, embedding

them into the forefoot and hindfoot zones of an open sandal to measure dynamic foot-ground interactions and ground reaction forces, establishing agreement against laboratory-grade instrumented treadmills[4].

Ozturksoy et al. constructed a multi-strap open sandal comprising a 3D-printed thermoplastic polyurethane base, pairing randomly routed optical fibers with six reference force-sensitive resistors to demonstrate that supervised learning regressors can decode complex footbed deformations without requiring an upper shoe counter[3].

Proulx et al. targeted low-resource clinical settings by embedding an articulated 704-cell piezoresistive sensor array into a flexible sandal base capable of withstanding dynamic forces up to 1,000 pounds while providing continuous pedobarographic visualization[4].

The domestic setting has likewise received empirical evaluation, recognizing that in South Asian households, footwear is systematically removed indoors, leaving neuropathic individuals exposed to unmonitored plantar trauma[3]. Malarvizhi et al. introduced an instrumented domestic slipper that integrates force transducers alongside calibrated temperature probes, continuously evaluating focal plantar loading and contralateral thermal discrepancies ($\Delta T \geq 2.2^{\circ}\mathrm{C}$) to generate automated prophylactic warnings before skin necrosis occurs[3].

Outside academia, commercial platforms such as SensorixFoot developed an open-toe smart sandal integrating baropodometric grids, thermal arrays, therapeutic vibratory mechanoreceptors, and hermetically sealed IP68 wireless charging modules for remote diabetic foot monitoring[3].

Similarly, decoupled smart textiles, including the Siren Care and Sensoria Health systems, embed sensors directly into elastomeric socks, completely bypassing the geometry of the shoe upper and enabling continuous plantar data acquisition in sandals, slippers, or during barefoot walking[1].

Indian national laboratories and clinical institutions have actively addressed the footwear requirements of the domestic diabetic population[3]. Between 70% and 87% of individuals with diabetes in India rely on open sandals, slippers, or chappals, as tropical heat and monsoonal moisture make closed shoes intolerable and induce fungal infections that accelerate skin breakdown[3].

In response, the Council of Scientific and Industrial Research, via the Central Leather Research Institute (CSIR-CLRI) in Chennai, partnered with the Prof. M. Viswanathan Diabetes Research Centre to engineer DIASTEP, a therapeutic open sandal granted Indian Patent No. 302551 in October 2018 and subsequently licensed for commercial manufacturing[3].

DIASTEP features a multi-density polyurethane unit sole with extended vertical depth to accommodate shock-absorbing elastomers, an integrated rocker-bottom sole profile that reduces propulsive forefoot pressures by guiding sagittal rollover, and cow softy leather retention straps engineered to stabilize the foot against lateral translational shear without causing friction blisters[3].

At the Indian Institute of Science (IISc) Bengaluru, a multidisciplinary bioengineering team led by Prof. G.K. Ananthasuresh and Dr. Pavan Belehalli replaced delicate electronic sensors and battery assemblies with passive mechanical metamaterials, publishing their results in *Wearable*

Technologies[3].

The IISc platform incorporates additively manufactured thermoplastic polyurethane footbeds embedded with compliant bistable snap-through arches[3]. Each arch is computationally dimensioned according to patient mass, foot geometry, and walking speed; when dynamic plantar stresses exceed 200 kPa—the clinical threshold for capillary occlusion and tissue ischemia—the arches snap into a negative-stiffness regime, offloading the vulnerable site and redistributing force to adjacent healthy tissue[3].

Upon dynamic unloading during the swing phase, the arches passively recover their undeformed baseline geometry[3]. In clinical evaluations involving neuropathic diabetic patients, this non-electronic system achieved a 57% peak dynamic pressure reduction across the forefoot[3].

At the domestic academic level, Chawre et al. designed an instrumented diabetic foot care sandal embedding flexible piezoresistive sensors beneath the calcaneus, metatarsal heads, and hallux, transmitting baropodometric signals into a LabVIEW acquisition module to trigger overpressure alarms[3].

The collective empirical evidence demonstrates that instrumenting open footwear, mitigating sandal-specific mechanical stresses, and developing solutions for the Indian diabetic cohort are established areas of clinical and biomedical engineering inquiry[3].

2. Biomechanical and Transducer Dynamics: Critical Flaws in the Simplified Sensor Sandal Concept

The technical premise underpinning early-stage proposals like Sole Sense—specifically, that embedding three to five discrete, single-axis Force Sensitive Resistors (FSRs) into a sandal footbed and acquiring data via an ESP32 microcontroller provides a viable clinical gait or diabetic foot ulcer screening tool—suffers from significant transducer-level and biomechanical oversights[1].

Diabetic peripheral neuropathy strips the lower extremity of protective nociception, allowing repetitive mechanical stresses to induce microvascular collapse, tissue necrosis, and full-thickness ulceration[3]. However, clinical bioengineering literature demonstrates that tissue breakdown is not governed solely by perpendicular compressive pressure, but by the multi-axial stress tensor operating at the dermis interface[3].

Planar shear stresses, acting along the anteroposterior and mediolateral axes, generate destructive internal tissue deformation[26]. Biomechanical evaluations by Yavuz et al. revealed that peak plantar shear stress and the shear-time integral are significantly elevated in diabetic neuropathic cohorts compared to non-neuropathic diabetic controls and healthy individuals[26]. Crucially, clinical mapping confirms that the anatomical site of peak shear stress frequently aligns with the precise location of past or emerging ulceration, whereas peak normal pressure often deviates from the breakdown locus[26].

$$\sigma_z = \frac{F_z}{A}$$

$$\tau = \frac{\sqrt{F_x^2 + F_y^2}}{A}$$

$$PTI = \int_{t_0}^{t_1} \sigma_z(t)\, dt \approx \sum_{i=1}^{N} \sigma_{z,i}\, \Delta t$$

$$STI = \int_{t_0}^{t_1} \tau(t)\, dt \approx \sum_{i=1}^{N} \tau_i\, \Delta t$$

In an enclosed orthopedic shoe, the rigid heel counter, deep quarter panels, and dorsal lacing mechanically constrain the tarsal and metatarsal skeleton, transferring dynamic deceleration and propulsive forces across the footwear upper[3].
In an open sandal, chappal, or slide, dorsal retention is governed by unpadded straps, leaving the foot unconstrained in the transverse plane[3]. During terminal stance and push-off, the plantar aspect of the foot slides across the footbed material, generating large tangential scrubbing forces[3].

Single-axis vertical FSRs are insensitive to orthogonal force components (F_x and F_u)[1]. By measuring only normal vertical pressure, a sparse FSR array remains blind to planar shear stress, omitting the primary physical driver of tissue ulceration in non-enclosed footwear[3].
The structural difference between footwear classes directly alters sensor registration[3].
Standard smart insoles rely on boundary constraints to maintain spatial registration, ensuring that individual sensing nodes stay aligned beneath specific anatomical landmarks such as the medial heel or the first metatarsal head[3].
In an open sandal, the lack of lateral containment permits significant dynamic translation and rotation of the foot relative to the sole[3]. During unstructured ambulation, the foot routinely drifts 8 to 12 millimeters across the footbed[3].

In a sparse four-sensor configuration, this translational drift causes the narrow load column beneath the metatarsal head to slide off the active area of the sensor entirely, impinging instead on the adjacent non-instrumented footbed elastomer[23].
The resulting data shows a sharp reduction in recorded voltage, which an onboard algorithm misclassifies as successful offloading or improved bilateral symmetry, even as the patient subjects insensate tissue to focal trauma immediately adjacent to the inactive sensor perimeter[3].



Transducer physics further limits the utility of standard polymer thick-film FSRs in clinical applications[1]. Piezoresistive FSRs exhibit substantial hysteresis (often between 15% and 25%), meaning that identical mechanical forces yield divergent electrical resistance values depending on whether the sensor is loading or unloading[1].
Under continuous static load, these sensors suffer from viscoelastic creep and baseline resistance drift, preventing accurate quantitative calibration in kilopascals without complex, time-dependent compensation algorithms[1].
Furthermore, when subjected to the repetitive multi-directional scrubbing characteristic of an

unconstrained sandal footbed, the thin polymer adhesive layers of standard FSRs undergo mechanical delamination[3]. The lateral sliding distorts the microscopic conductive pathways between the flexible substrate and the ink layer, causing calibration drift and electrical open circuits within several thousand steps[1].

A parallel thermodynamic challenge undermines simplistic implementations of open-footwear thermometry[4]. In an enclosed shoe, the internal void functions as a quasi-insulated thermodynamic chamber that dampens exterior fluctuations, establishing a stable microclimate where bilateral temperature differentials ($\Delta T \geq 2.2^{\circ}\mathrm{C}$) reliably indicate subcutaneous inflammation[3].

An open sandal exposes the plantar dermis directly to turbulent ambient convection, evaporative cooling from sweat dissipation, and direct solar radiation[4]. A localized rise in tissue temperature can be masked by an ambient breeze, while unshielded sunlight can induce an artificial thermal spike[4].

Without dynamic differential calibration algorithms that normalize plantar readings against non-weight-bearing dorsal reference sensors and ambient thermistors, open-air thermography yields frequent false positives and false negatives[4].

3. Formulation of Validated, High-Impact Research White Spaces

Because the general concept of instrumenting sandals is extensively documented, academic researchers and technology developers must abandon generic claims of open-footwear novelty[1]. Instead, engineering efforts should target four authentic, unresolved research white spaces spanning transducer physics, environmental materials science, cultural biomechanics, and spatial machine learning algorithms.

Fresh White Space A: Triaxial Plantar Load Decoupling & Transduction in Non-Enclosed Footwear Interfaces

While complex in-shoe multi-axis systems—such as the 64-node inductive SLIPS array or optical digital image correlation insole scaffolds—have demonstrated triaxial capabilities in controlled laboratory shoes, no system has successfully resolved multi-axial forces within an open sandal[28].

The primary white space lies in developing ultra-low-profile, flexible sensing matrices capable of mechanically and electronically decoupling normal compressive pressure (σ_z) from orthogonal planar shear stresses (τ_{xz}, τ_{uz}) directly beneath the unconstrained foot[28].

Triaxial Transduction and Processing Topology



Resolving this white space requires engineering multi-layered elastomeric sensor matrices (utilizing micro-structured polydimethylsiloxane or compliant polyurethane formulations) that measure lateral shear strain independently of vertical displacement[30].

Such devices must extract the instantaneous Shear-Time Integral alongside the Pressure-Time Integral under loose strap retention, providing the clinical community with the first quantitative data defining the precise thresholds where tangential friction triggers tissue breakdown in tropical climates[26].

Fresh White Space B: Multi-Modal Ingress Robustness and Accelerated Cyclic Degradation Under Tropical Monsoonal Conditions

Wearable gait sensors are predominantly evaluated within clean, dry, climate-controlled biomechanics laboratories[1]. No published investigation has characterized the electromechanical durability, transducer drift, and physical degradation mechanisms of flexible sensor footbeds exposed to the environmental extremes of South Asian and equatorial environments[1].

An authentic research gap exists in evaluating sensorized open footwear under dynamic fluid and particulate ingress, including high-salinity water immersion, road mud contamination, and abrasive silica dust exposure during active mechanical flexing[1].

Engineering research must characterize transducer drift, dielectric breakdown, and delamination across more than 500,000 continuous gait cycles within environmental test chambers maintained at $35^{\circ}\mathrm{C}–45^{\circ}\mathrm{C}$ and 85%–95% relative humidity[1].

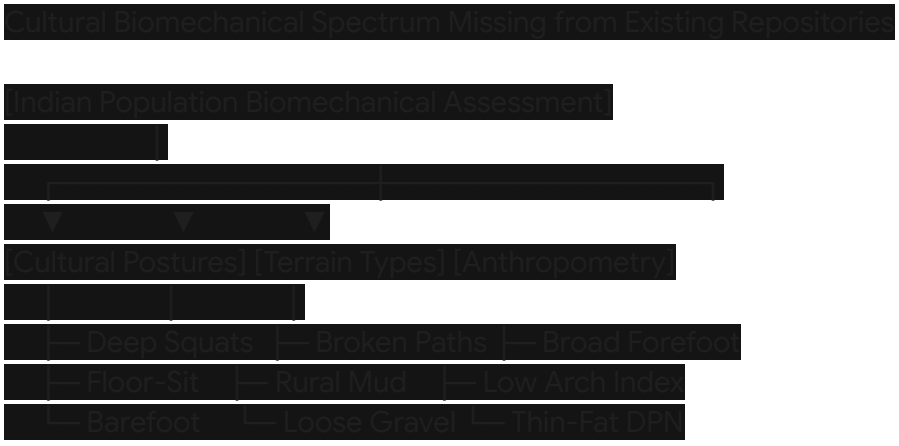
This requires establishing viable hermetic encapsulation methods—such as parylene-C chemical vapor deposition, flexible polyurethane potting, and sealed inductive charging

interfaces—that protect electrical pathways without creating rigid, localized pressure points that could induce ulcers on neuropathic feet[1].

Fresh White Space C: Open-Access Biomechanical Datasets of Culturally Specific Indian Postures and Unstructured Ambulation

Publicly available smart insole and gait databases (such as Insole-GAITRite, ImbalanSense, and WearGait-PD) are derived exclusively from Western or East Asian cohorts performing level, obstruction-free walking across indoor walkways[2].

No open-access, high-density plantar pressure and kinetic database captures the unique anthropometric features, gait patterns, and cultural postures of the Indian population[1].



A significant research contribution involves capturing synchronized, ground-truth kinetic and kinematic datasets during cultural activities, including deep full squatting (associated with rural chores and squat toilet utilization), cross-legged floor sitting (*Sukhasana*), load carrying, and transition dynamics across unstructured surfaces such as broken concrete, loose gravel, and mud[1].

Furthermore, datasets must document plantar stress distributions across Indian foot morphologies, which feature broader forefoot-to-rearfoot width ratios, lower static longitudinal arch indices, and the characteristic "thin-fat" diabetic phenotype where severe metabolic neuropathy develops at lower Body Mass Index thresholds than in Western populations[1].

Publishing these repositories under open-science frameworks will provide the benchmark data required to train and evaluate generalized edge-AI models[1].

Fresh White Space D: Edge-AI Spatial Invariance and Misregistration-Resilient Kinematic Estimation

Current wearable gait algorithms rely on static spatial mapping, assuming that a given transducer channel maps reliably to an invariant anatomical landmark[23]. In open footwear,

where unconstrained strap dynamics induce continuous planar foot translation, standard machine learning pipelines experience marked signal degradation and feature mismatch[1].
Developing edge-AI architectures specifically designed to achieve spatial invariance represents a critical, unfilled technical niche[1]. Rather than treating sensor inputs as fixed temporal channels, models must incorporate Graph Neural Networks (GNNs) or Spatial Transformer layers that treat sensor nodes as dynamic topological graphs[1].
This allows the network to estimate the true Center of Pressure (CoP) trajectory and temporal gait phases independently of global translational shifts across the footbed[1].
Furthermore, replacing sparse discrete arrays with distributed sensing media (such as embedded optical fiber grids or conductive textile meshes) enables models to decode bulk structural deformation rather than relying on point-source contact[3].
These models must be compressed via 8-bit integer quantization (INT8) to execute real-time inference locally on low-power microcontrollers (such as the ESP32-S3 or ARM Cortex-M33) within a sub-$50\ \mathrm{mW}$ power budget[1].



The matrix below contrasts the refuted initial claims against the verified prior art and the newly formulated research white spaces.

Analysis	Refuted Initial Claim (Sole Sense	Verified State of	Authentic Research White

Dimension	Concept)	Prior Art	Space
Footwear Carrier Novelty	No sensor, academic paper, or product has ever embedded electronics into open footwear[1].	Multiple peer-reviewed sensorized sandals (Safarloo 2022, Burch 2023, Ozturksoy 2020) and commercial devices (SensorixFoot)[3].	**Triaxial Force Decoupling:** Isolating normal vertical pressure from planar shear stresses under unconstrained sandal straps[28].
Indian Diabetic Coverage	Indian diabetic patients wearing open sandals have been completely ignored by researchers[1].	CSIR-CLRI DIASTEP patent (No. 302551); IISc snapping arch sandals (57% forefoot reduction); Chawre et al. sensor sandal[3].	**Environmental Resilience:** Transducer packaging and hermetic potting surviving tropical monsoonal water, mud, and dust (>500k cycles)[1].
Transducer Modality	3 to 5 discrete vertical FSRs provide adequate data for diabetic foot ulcer screening[23].	Standard FSRs exhibit 15%–25% hysteresis, severe creep, and cannot detect destructive shear stresses (F_x, F_u)[1].	**Continuous Metrology:** Distributed sensing meshes and flexible nanocomposites that mitigate discrete sensor delamination[3].
Biomechanical Registration	Foot position over footbed sensors is assumed static and equivalent to enclosed shoes[23].	Unconstrained footbeds allow 8 to 12 mm of multi-axis translation, generating spatial misregistration and false offloading signals[3].	**Spatially Invariant Edge-AI:** Graph neural networks that track true Center of Pressure trajectories despite dynamic foot sliding[1].
Plantar	Closed-shoe contralateral	Open-air convective airflow,	**Multi-Site Differential**

Thermography	temperature monitoring ($\Delta T \geq 2.2^{\circ}\mathrm{C}$) applies directly to open sandals[1].	evaporative sweat dissipation, and solar irradiance invalidate uncompensated thermal differentials[4].	**Algorithms:** Normalizing plantar readings against dorsal reference probes and ambient environmental thermistors[4].
Dataset Availability	Indian gait data is missing simply because nobody has engineered a local insole[1].	Insole gait datasets exist globally, but entirely lack non-Western cultural postures and Indian foot anthropometry[1].	**Culturally Stratified Gait Database:** Curating open-access repositories covering deep squats, floor sitting, and unstructured terrain[1].

4. Regulatory Engineering, Clinical Validation Protocols, and Translational Roadmaps

Transitioning an open-footwear sensing platform from an academic concept into a certified medical device demands a structured regulatory strategy, rigorous benchmarking against clinical gold standards, and coordinated grant deployment[1].

CDSCO Regulatory Classification and Licensing Pathway in India

In India, medical devices are regulated by the Central Drugs Standard Control Organisation (CDSCO) under the *Medical Devices Rules, 2017*[1]. Devices are categorized into four risk classes: Class A (low risk), Class B (low-to-moderate risk), Class C (moderate-to-high risk), and Class D (high risk)[1].

The intended use and marketing claims dictate the regulatory pathway[1]. If an instrumented sandal is registered strictly as a general wellness mobility tracker, non-invasive gait trainer, or physical rehabilitation accessory that provides basic step metrics without clinical claims, it falls under Class A (non-measuring)[1]. This route requires basic registration on the CDSCO SUGAM online portal, exempting the sponsor from extensive premarket clinical audits[1].

Conversely, if the device claims to measure quantitative plantar pressure in kilopascals, screen for diabetic peripheral neuropathy complications, provide diagnostic alerts for impending ulcers, or guide clinical offloading interventions, it is classified as an Active Diagnostic Medical Device falling under **Class B**[1].



Securing a Class B manufacturing license requires the applicant to submit Form MD-3 to the State Licensing Authority via the SUGAM portal[35]. The manufacturing establishment must comply with the Fifth Schedule of the Medical Devices Rules, 2017, and establish an ISO 13485:2016 certified Quality Management System[35].

The technical dossier must include a comprehensive Device Master File, incorporating electrical safety testing under IEC 60601-1, electromagnetic compatibility testing under IEC 60601-1-2, skin-contact biocompatibility evaluations under ISO 10993 (cytotoxicity, sensitization, and irritation), software lifecycle documentation under IEC 62304, and formal risk management files under ISO 14971[35].

Following technical submission, an accredited Notified Body conducts an on-site facility audit within 90 days; upon resolution of inspection findings, the State Licensing Authority issues the commercial manufacturing license on Form MD-5[35].

For international market entry, alignment with United States Food and Drug Administration (FDA) classifications is required[1]. Gait analysis systems typically fall under Regulation 21 CFR 890.5360 (Measuring device, kinetic and kinematic parameters), classified as Class II devices under Product Code LXH[45]. Plantar pressure mapping systems fall under Product Code NXE[45]. Certain movement-analysis platforms, such as Stappone, have achieved market access through 510(k)-exempt provisions by restricting their intended use to recording objective gait parameters and temporal symmetry without offering autonomous diagnostic claims[1].

However, if an instrumented sandal introduces autonomous machine learning algorithms to

predict tissue ulceration, the FDA requires a full 510(k) premarket notification demonstrating substantial equivalence to clinical predicate devices, such as the Tekscan F-Scan or Novel Pedar-X systems[1].

Gold-Standard Clinical Validation Protocol

To establish clinical validity, any prototype instrumented footwear system must undergo concurrent validation against accepted laboratory reference platforms[1].



The experimental benchmarking protocol must incorporate both in-shoe reference matrices and ground-based kinetic platforms[1]. In-footwear pressure distribution is measured simultaneously using a calibrated Novel Pedar-X capacitive matrix or a Tekscan F-Scan piezoresistive grid positioned directly at the plantar-footwear interface[1].
Simultaneously, participants ambulate across synchronized multi-axis laboratory force plates (such as AMTI or Kistler platforms) while tracking retroreflective markers using multi-camera optoelectronic motion analysis systems (Vicon or Qualisys) to capture ground reaction forces (F_x, F_y, F_z) and joint kinematics[1]. In India, accessible validation infrastructure includes the indigenous multi-axial force plate systems engineered at the National Institute of Technology (NIT) Rourkela[2].

The evaluation requires specific statistical benchmarks:

- Concurrent validity and test-retest reliability across peak pressure, contact duration, and the pressure-time integral must achieve an Intraclass Correlation Coefficient ($\mathrm{ICC}_{2,1}$) exceeding 0.85 across all anatomical masks[50].
- Bland-Altman 95% Limits of Agreement ($\text{Mean Bias} \pm 1.96 \cdot \text{SD}$) must be documented for distinct anatomical regions, specifically the medial heel, lateral heel, midfoot, metatarsal heads 1 through 5, and the hallux[10].
- The Root Mean Square Error (RMSE) for dynamic vertical force estimation throughout the complete stance phase must remain below 7% of total body weight[3].

Translational Grant Ecosystem and Academic Partnerships in India

Developing custom, environmentally robust medical hardware requires non-dilutive capital and clinical collaboration before commercial scaling[1]. In India, biomedical consortia can access several public funding schemes tailored to hardware development and clinical translation[1].

Funding Agency & Scheme	Funding Quantum	Target Technology Readiness Level	Strategic Mandate and Focus
DST NIDHI PRAYAS[1]	₹20 Lakhs to ₹40 Lakhs	TRL 1 to TRL 3 (Proof of Concept to Early Hardware)	Supports fabrication of physical prototypes, benchtop transducer evaluation, and proof-of-concept testing[1].
BIRAC BIG (Biotechnology Ignition Grant)[1]	Up to ₹50 Lakhs	TRL 2 to TRL 4 (Early Prototype to Functional System)	Translates medical technologies into validated functional prototypes; requires startup incorporation or incubator affiliation[1].
ICMR NIRMAN-AT[1]	Project-dependent	TRL 4 to TRL 6 (Clinical Validation & Field Trials)	Dedicated to developing, testing, and commercializing

			assistive technologies and rehabilitation hardware for Indian healthcare[1].
BIRAC PACE / SPARSH	₹30 Lakhs to ₹1 Crore	TRL 3 to TRL 5 (Academic-Industrial R&D)	Encourages translational partnerships addressing high-priority public health and affordable healthcare challenges.

To achieve clinical translation, engineering teams should establish formal collaborative consortia with specialized Indian institutions:

- **Clinical Trial Sites:** The Karnataka Institute of Endocrinology and Research (KIER, Bengaluru), the Prof. M. Viswanathan Diabetes Research Centre (MVDRC, Chennai), Christian Medical College (CMC, Vellore), and the All India Institute of Medical Sciences (AIIMS, New Delhi) provide access to high-volume diabetic foot clinics and institutional review boards[2].
- **Biomechanical Validation Facilities:** The CSIR-Central Leather Research Institute (CSIR-CLRI, Chennai), the Indian Institute of Science (IISc, Bengaluru), and the National Institute of Technology (NIT, Rourkela) host specialized footwear testing laboratories, ISO-compliant materials testing infrastructure, and pedobarographic validation equipment[2].





By transitioning away from generic claims of open-footwear novelty and addressing the true engineering hurdles of shear stress decoupling, monsoonal durability, cultural activity profiling, and spatial machine learning, bioengineering consortia can establish defensible clinical platforms capable of substantially improving diabetic foot care across India and tropical health systems globally[1].

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